PGR216: AI and Cybersecurity Applications

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Logistic regression for Multiclass classification

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Content

1. MultiClass Classificaton
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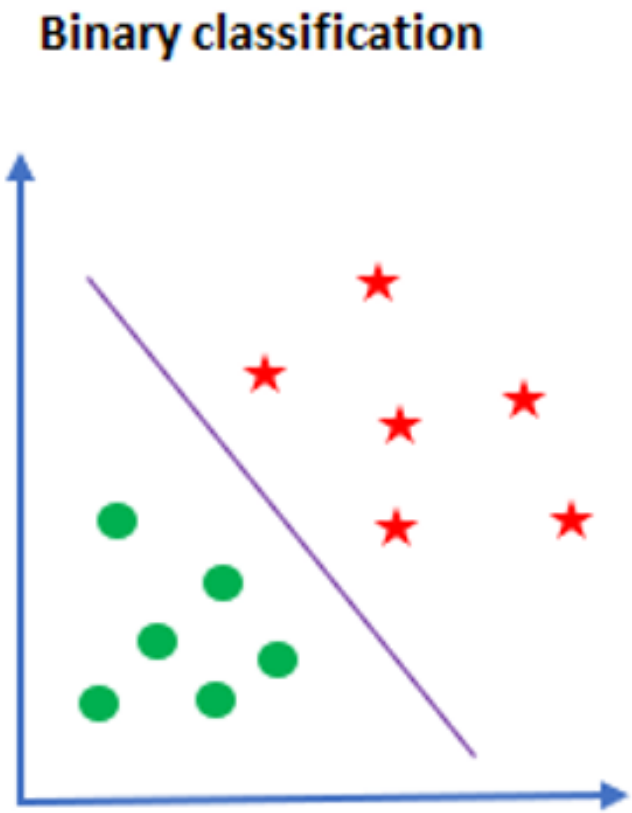
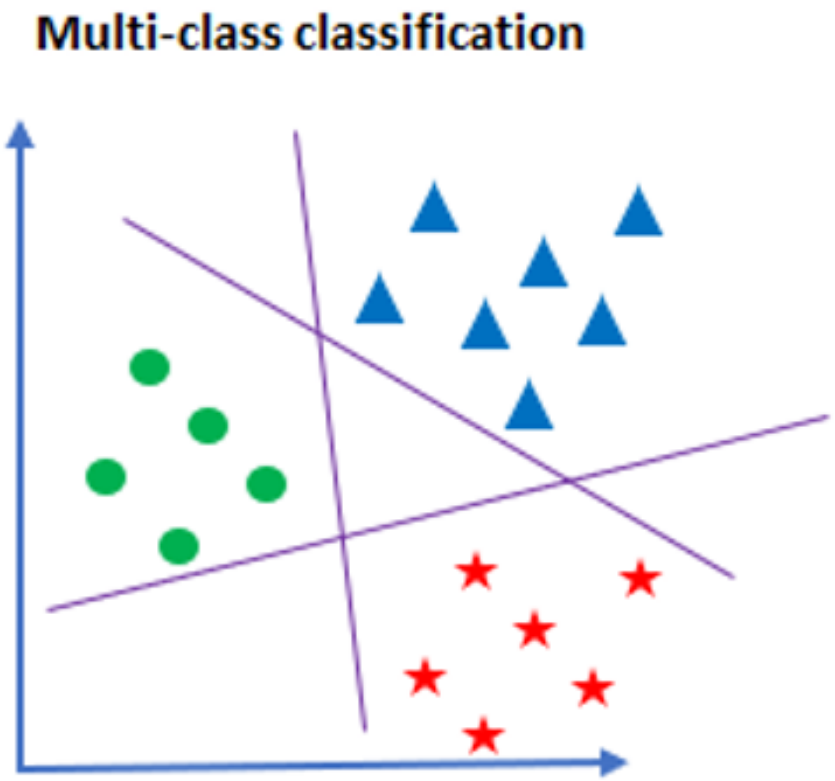
Binary Vs. MultiClass





The Classification Problem

We are given:

$$x \in \mathbb{R}^d, \quad y \in \{1, 2, \ldots, K\}$$

Goal:

- Learn a model that predicts the class of x
- Output probabilities:

$$P(y = k \mid x)$$

Questions:

- Logistic regression is binary... so how do we handle $K > 2$?
- Do we modify the model or use many models?

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Reminder: Binary Logistic Regression

For $y \in \{0, 1\}$:

$$P(y = 1 \mid x) = \sigma(w^T x)$$

where sigmoid:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Decision rule:

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1|x) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

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Loss Function (Binary Cross-Entropy)

Training minimizes:

$$L = -\left[y \log \sigma(w^T x) + (1 - y) \log(1 - \sigma(w^T x))\right]$$

This is called:

- Log-loss
- Cross-entropy loss

What Breaks for Multi-Class?

If $y \in \{1, 2, 3\}$:

- Sigmoid gives only ONE probability
- But we need THREE probabilities
- They must sum to 1:

$$\sum_{k=1}^{K} P(y = k \mid x) = 1$$

Sigmoid cannot do this.

Problem Setting

- Many classifiers are inherently **binary**
  - Logistic Regression
  - Support Vector Machines
  - Perceptron
- Real-world problems often involve **multiple classes**

$$y \in \{1, 2, \ldots, K\}$$

- Key question:

How can we extend binary classifiers to handle multi-class problems?

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Three Strategies

- **One-vs-Rest (OvR)** / One-vs-All
- **One-vs-One (OvO)**
- **Multiclass Model**

The first two strategies decompose a multi-class problem into multiple binary classification tasks. However, the third category comes up with a model that perform multiclass classification by design.

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One-vs-Rest (OvR): Idea

- Train **K binary classifiers** for a K-class problem
- Each classifier:
  - Positive class: one specific class
  - Negative class: all remaining classes

For class k:

$$y = \begin{cases} 1 & \text{if class } = k \\ 0 & \text{otherwise} \end{cases}$$

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One-vs-Rest: Training

For K classes:

- Train classifiers:

$$f_1(x), f_2(x), \ldots, f_K(x)$$

- Each classifier sees:
  - Many negative samples
  - Relatively few positive samples

This can lead to **class imbalance**.

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One-vs-Rest: Prediction

- Evaluate all K classifiers
- Choose class with highest confidence score:

$$\hat{y} = \arg\max_k f_k(x)$$

Important:

- Scores must be comparable
- Calibration matters (especially for SVMs)

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One-vs-Rest: Pros & Cons

Advantages

- Simple and intuitive
- Only K classifiers
- Scales well to many classes

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Problems with OvR

- Class imbalance
- Ambiguous regions (multiple classifiers may fire)
- Poor separation when classes overlap strongly
- Probabilities are not normalized
- Models trained independently
- No competition between classes
- Decision boundaries can be inconsistent

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One-vs-One (OvO): Idea

- Train a classifier for **every pair of classes**
- For K classes:
$$\frac{K(K-1)}{2} \text{ classifiers}$$
- Each classifier only sees samples from **two classes**

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One-vs-One: Training

For classes i and j:

$$y = \begin{cases} 1 & \text{if class } = i \\ 0 & \text{if class } = j \end{cases}$$

- Balanced datasets
- Simpler decision boundaries
- Faster training per classifier

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One-vs-One: Prediction

- Each classifier votes for a class
- Final decision via **majority voting**:

$$\hat{y} = \arg\max_k \text{votes}_k$$

Alternative:

- Weighted voting
- Probability coupling

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One-vs-One: Pros & Cons

Advantages

- Better performance for complex boundaries
- Balanced subproblems
- Works very well with SVMs

Disadvantages

- Many classifiers
- Slower inference
- Higher memory usage

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OvR vs OvO: Comparison

	OvR	**OvO**
# Classifiers	K	$K(K-1)/2$
Training Speed	Fast	Slower
Inference Speed	Fast	Slower
Class Imbalance	Yes	No
Boundary Quality	Moderate	Strong

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Direct Multiclass Models

Some models are **natively multi-class**:

- Softmax Logistic Regression
- Neural Networks
- Decision Trees

Softmax example:

$$P(y = k|x) = \frac{e^{w_k^T x}}{\sum_{j=1}^{K} e^{w_j^T x}}$$

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Multinomial Logistic Regression

We generalize sigmoid to **softmax**.
For each class k:

$$P(y = k \mid x) = \frac{e^{w_k^T x}}{\sum_{j=1}^{K} e^{w_j^T x}}$$

This is called:

- Multinomial Logistic Regression
- Softmax Regression
- Maximum Entropy Classifier

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Why Softmax Works

Softmax guarantees:

$$\sum_{k=1}^{K} P(y = k \mid x) = 1$$

All classes are learned **jointly**.
Classes compete during training.

Connection to Binary Logistic Regression

For $K = 2$:

$$P(y = 1|x) = \frac{e^{w_1^T x}}{e^{w_1^T x} + e^{w_2^T x}}$$

This simplifies to the sigmoid form.

Binary logistic regression is a special case of softmax.

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Model Parameters

We now have:

$$W = \begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_K^T \end{bmatrix}$$

Each class has its own weight vector.

Loss Function: Categorical Cross-Entropy

Using one-hot encoding y_k:

$$L = -\sum_{k=1}^{K} y_k \log P(y = k \mid x)$$

This is the same loss used in neural networks.

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Decision Rule

$$\hat{y} = \arg\max_k P(y = k \mid x)$$

This corresponds to:

$$\hat{y} = \arg\max_k w_k^T x$$

So decision boundaries are linear.

Geometric Interpretation

- Each class defines a linear score $w_k^T x$
- The class with highest score wins
- Space is partitioned into K linear regions

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OvR vs Multinomial

	OvR	Multinomial
Models	K	1
Probabilities sum to 1	No	Yes
Joint training	No	Yes
Theoretically correct	No	Yes
Performance	Good	Better

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Why Multinomial is Preferred

- Proper probabilistic model
- Better calibrated probabilities
- More stable boundaries
- Scales better with many classes

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Big Insight

$$\text{Logistic Regression} + \text{Softmax} = \text{Multi-Class Logistic Regression}$$

Same idea. Different activation.

Connection to Neural Networks

A neural network final layer:

$$z = Wx, \quad \hat{y} = \text{softmax}(z)$$

This is exactly multinomial logistic regression.
Logistic regression and multinomial regression forms the fundamental building block of Neural networks.

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Other Possibility: Hierarchical Classification

- Classes arranged in a hierarchy
- Binary decisions at each node

Example:

- Animal $\rightarrow$ Mammal $\rightarrow$ Dog
- Animal $\rightarrow$ Bird $\rightarrow$ Sparrow

Efficient for very large class sets.

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Practical Guidelines

- Few classes ($K < 10$): OvO often performs best
- Many classes: OvR scales better
- Large datasets + deep learning: native multi-class
- Need robustness: consider ECOC

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Summary

- Logistic regression is not only binary, could be extended to multiclass problems.
- OvR is a workaround
- OvR: simple, scalable, but imbalanced
- OvO: accurate, but computationally expensive
- Softmax gives the true formulation
- Same loss as deep learning
  - Linear decision boundaries **Choice depends on data, model, and constraints**

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